

SMC Trading Bot

Complete Specification Document

A Telegram-controlled trading bot using APA (Advanced Price Action) and Supply & Demand strategies with multi-factor scoring, backtesting, and auto-execution.

■■ This is a configurable rule engine, not a guaranteed profitable system. Trading involves substantial risk of loss. Always test in paper mode first.

Overview

The bot analyzes currency pairs, synthetic indices, and gold using professional APA/SMC concepts. It scores each setup using a 7-factor system, and auto-executes trades that pass a configurable threshold with a minimum 1:3 risk-reward ratio. Includes full backtesting, entry confirmation, trade management, session filtering, and news filtering.

Architecture

Module	Description
analysis/structure.py	APA market structure: BOS, CHoCH, OB, FVG, liquidity pools
analysis/supply_demand.py	S/D zone detection with freshness and strength scoring
analysis/scoring.py	7-factor signal scoring engine
analysis/confirmation.py	Entry confirmation: zone retest + candle patterns
analysis/sessions.py	Session filtering (London, NY, Tokyo)
data/provider.py	Real market data: MT5 → Twelve Data → yfinance → synthetic
executors/paper.py	Paper trading simulator (default)
executors/mt5.py	MetaTrader 5 live execution
execution/manager.py	Trade management: breakeven, trailing, partial close
risk/manager.py	10 hard risk gates + position sizing
news/filter.py	Economic calendar news filter (ForexFactory)
backtest/engine.py	Full backtesting engine with metrics
backtest/runner.py	CLI + Telegram backtest runner
bot/handlers.py	Telegram command handlers
storage/db.py	SQLite persistence
scheduler.py	Market scanner + auto-execution loop

APA (Advanced Price Action) Analysis

Component	Description
Swing Detection	Identifies swing highs/lows using configurable lookback (default 3 bars)
Market Structure	Bullish (HH+HL), Bearish (LH+LL), or Ranging
BOS	Break of Structure — trend continuation signal
CHoCH	Change of Character — trend reversal signal (tracks prior trend from 3rd swing)
Order Blocks	Last opposite candle before institutional impulse move
Fair Value Gaps	3-candle imbalances where price hasn't returned
Liquidity Pools	Equal highs (buy-side) and equal lows (sell-side) where stops accumulate
Premium/Discount	Equilibrium-based zone classification

Supply & Demand Zones

Pattern	Type	Description
Rally-Base-Drop	Supply	Up → consolidation → sharp drop
Drop-Base-Drop	Supply	Down → consolidation → sharp drop
Drop-Base-Rally	Demand	Down → consolidation → sharp rally
Rally-Base-Rally	Demand	Up → consolidation → sharp rally

Each zone is scored on impulse strength (40%), freshness (40%), and base tightness (20%). Fresh zones have not been revisited by price since creation.

7-Factor Scoring System

#	Factor	Weight	Max
1	Market Structure Alignment	20%	20
2	Supply/Demand Zone	15%	15
3	Order Block Confluence	15%	15
4	Fair Value Gap	10%	10
5	Liquidity Sweep	15%	15
6	Risk-Reward Ratio	15%	15
7	MTF Confluence	10%	10
	Total	100%	100

Entry Confirmation

Before executing, the bot verifies entry confirmation (configurable):

Confirmation	Description
Zone Retest	Price must have touched the S/D zone within recent bars
Engulfing	Bullish/bearish engulfing candle pattern
Pin Bar	Hammer/shooting star with long wick
Inside Bar Breakout	Mother candle → inside bar → breakout in trade direction
Displacement	Strong institutional impulse (body > 2×ATR)

Risk Management — 10 Hard Gates

Gate	Description	Default
Auto-trade enabled	auto_trade=true and not paused	OFF
Symbol allowed	Symbol is in configured list	—
Symbol cooldown	No trade within cooldown period	30 min
Daily loss limit	PnL > -(balance × max_loss%)	5%
Daily trade count	Trades today < max	10
Max positions	Open positions < max	5
Score threshold	Score ≥ threshold	40%
Min RR ratio	RR ≥ min_rr_ratio	1:3
Spread check	Spread ≤ max_spread_pips	5 pips
Free margin	Free margin > 2× required	—

Trade Management

Once a trade is open, the bot actively manages it:

Feature	Description	Default
Breakeven	Move SL to entry at 1R profit	1.0R
Partial Close	Close 50% at 2R, trail the rest	2.0R / 50%
Trailing Stop	ATR-based trailing behind price	2×ATR
Time Exit	Close if held >100 bars without 1R	100 bars

Session & News Filtering

The bot only trades during high-liquidity sessions:

Session	Hours (UTC)	Default
Tokyo	00:00 - 09:00	OFF
London	08:00 - 17:00	ON
New York	13:00 - 22:00	ON
London/NY Overlap	13:00 - 17:00	ON

News Filter: Uses ForexFactory economic calendar (free, no API key). Blocks trading 15 minutes before and after high-impact events for the relevant currencies.

Backtesting Engine

Full historical replay through the analysis pipeline with realistic spread, slippage, and commission.

Run from Telegram: /backtest EURUSD H1 180
Run from CLI: python -m backtest.runner --symbol EURUSD --timeframe H1 --days 180

Metrics computed:

Metric	Description
Win Rate	Percentage of winning trades
Profit Factor	Gross profit / gross loss
Expectancy	Average P&L per trade
Max Drawdown	Largest peak-to-trough decline (%)
Sharpe Ratio	Risk-adjusted return (annualized)
Sortino Ratio	Downside-adjusted return
Monthly Returns	P&L breakdown by month
Avg RR	Average risk-reward achieved
Avg Bars Held	Average trade duration

Real Market Data Sources

Paper mode now uses real market data instead of synthetic random walks:

Source	Instruments	Cost
MetaTrader 5	All (forex, gold, synthetic indices)	Free (needs MT5 terminal)
Twelve Data API	Forex + gold (140 currencies)	Free tier: 8 req/min, 800/day
yfinance	Forex + gold	Free, no API key
Synthetic	Synthetic indices (fallback)	Free (random walk)

Telegram Commands

Command	Description
/start	Main menu with quick stats
/scan	Scan all symbols for signals
/analyze [symbol]	Deep analysis of specific symbol
/backtest [symbol] [tf] [days]	Run historical backtest
/positions	Show open positions
/close_all	Close all positions (with confirmation)
/settings	Adjust all settings via inline keyboards
/account	Show account info
/history	Recent trade history
/sessions	Show trading session status
/news [symbol]	Check news blackout status
/pause / /resume	Pause/resume auto-trading
/mode [paper live]	Switch execution mode
/risk [pct]	Set risk per trade
/rr [ratio]	Set min RR ratio
/score [val]	Set score threshold

Tech Stack

Component	Technology
Language	Python 3.11+
Telegram	python-telegram-bot v21
Scheduling	APScheduler
Data Analysis	pandas, numpy
MT5 Integration	MetaTrader5 Python package
Market Data	Twelve Data API, yfinance
Database	SQLite (aiosqlite)
Backtesting	Custom engine with full metrics
Containerization	Docker, docker-compose

Safety Defaults

Setting	Default	Rationale
TRADING_MODE	paper	No real money at risk until user opts in
AUTO_TRADE	false	User must explicitly enable
RISK_PER_TRADE	1.0%	Conservative default
MAX_DAILY_LOSS_PCT	5.0%	Stop trading after 5% daily loss
SCORE_THRESHOLD	40.0%	Configurable threshold
MIN_RR_RATIO	3.0	1:3 minimum as requested
Entry Confirmation	ON	Zone retest + candle pattern required
Session Filter	London+NY	Only trade high-liquidity sessions
News Filter	ON	Block during high-impact news